

Hasnain Abbas

Experimental physicist · Memristive & synaptic devices for neuromorphic computing

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RESEARCH PROFILE

MPhil physics researcher at Quaid-i-Azam University working across the full memristive stack: from sol-gel synthesis of Co-substituted BiFeO₃ thin films, through ferroelectric and resistive-switching characterization, to device-aware neural-network simulation that uses measured device conductance as synaptic weights. Seeking a PhD position in **neuromorphic computing**, **synaptic devices**, and **memristor-based hardware**.

RESEARCH INTERESTS

- Memristive & synaptic devices
- Neuromorphic & in-memory computing hardware
- Resistive switching in oxide thin films
- Ferroelectric & multiferroic thin films
- Device-aware neural-network simulation
- Sol-gel processing of functional ceramics

EDUCATION

- Quaid-i-Azam University Islamabad

Master of Philosophy (MPhil) in Physics

Islamabad, Pakistan

Sep 2024 – Present (Expected Sep 2026)

 - Thesis:** Co-substituted BiFeO₃ Thin Films for Neuromorphic Computing and Memristor Applications
 - Supervisor:** Dr. Ghulam Hassnain Jaffari
- COMSATS University Islamabad

Bachelor of Science (BS) in Physics, CGPA: 3.2/4.0 (79.9%)

Islamabad, Pakistan

Feb 2019 – Feb 2023

 - Relevant coursework:** Solid State Physics, Statistical Physics, Quantum Optics, High Energy Physics, Biophysics
 - Thesis:** Fast Forward Stimulated Raman Adiabatic Transfer of Atomic Population in a Delta-Type Atomic Medium (analytical modeling)
- FG Boys Inter College Karachi Cantt

Higher Secondary School Certificate (HSSC)

Karachi, Pakistan

Apr 2016 – Aug 2018

RESEARCH EXPERIENCE

- Quaid-i-Azam University Islamabad

Graduate Researcher, Thin-Film and Memristor Lab

Islamabad, Pakistan

Sep 2024 – Present



- Fabrication:** Synthesize Co-substituted BiFeO₃ ceramic and polymer-blended thin films by sol-gel spin coating, tuning precursor chemistry, annealing profile, and film thickness to obtain stable resistive switching
- Structural characterization:** Verify phase purity and microstructure by XRD, FTIR, thickness profilometry, and impedance spectroscopy, and correlate these with the measured switching response
- Ferroelectric characterization:** Extract remanent polarization and coercive field across compositions from P-E hysteresis loops on a Sawyer-Tower ferroelectric tester
- Memristive / synaptic testing:** Automate a Keithley 2602 SourceMeter with custom Lua (TSP) scripts for reproducible DC sweep and pulse protocols, producing high-resolution I-V and conductance data for resistive-switching characterization
- Device-to-system benchmarking:** Map experimentally extracted device conductance into the CrossSim neuromorphic simulator and train MLP networks in PyTorch on MNIST, benchmarking inference accuracy with the measured devices acting as synaptic weights

- Pakistan Institute of Nuclear Science and Technology (PINSTECH)

Research Intern, Optics and LINAC Labs

Islamabad, Pakistan

- Synthesized colloidal silver and gold nanoparticles by pulsed laser ablation
- Generated and analyzed shock waves driven by a high-power laser system
- Rotated through the Optics Lab, the LINAC facility, and specialized high-energy physics labs

TECHNICAL SKILLS

Thin-film & device fabrication	Sol-gel synthesis of ceramic and polymer-blended films, spin coating, thermal annealing, pulsed laser ablation (Ag and Au nanoparticles)
Electrical & memristive testing	Keithley 2602 SourceMeter (TSP scripting), DC sweep and pulse protocols, I-V and resistive-switching measurements, impedance analysis
Materials characterization	XRD, FTIR, Sawyer-Tower ferroelectric tester (P-E loops), thickness profilometry
Neuromorphic simulation & ML	CrossSim crossbar simulator, device-aware MLP training and evaluation on MNIST in PyTorch
Programming	Python (PyTorch, NumPy, Matplotlib), Lua (Keithley TSP), C++, Mathematica
Other software	LaTeX, Adobe Creative Suite (scientific figures, image and video editing)
Languages	English (fluent), Urdu (fluent), Balti (native)

TEACHING EXPERIENCE

FG Boys Degree College Skardu Skardu, Gilgit-Baltistan, Pakistan
Lecturer in Physics Mar 2023 – Aug 2024

- Taught BS Physics coursework and mentored students through problem sessions across three academic terms
- Supervised higher-secondary laboratory sessions and demonstrated core experiments

Gilgit Baltistan Institute of Science and Technology Skardu, Pakistan
Visiting Lecturer Oct 2023 – Dec 2023

- Lectured and supervised laboratory work for undergraduate medical students, linking physics principles to medical applications

CONFERENCES & WORKSHOPS

- 48th International Nathiagali Summer College on Physics and Contemporary Needs, Nathiagali, Pakistan (Jul 2023)
- Capacity Building Sessions for Lecturers, FG Boys Degree College Skardu, HEC Pakistan (Apr 2023)
- Workshop on Fundamentals of Quantum Mechanics, PIEAS, Islamabad (Jan 2023)

AWARDS & HONORS

- Best Lecturer Award, Al-Zehra Girls College, Skardu (Dec 2023)
- Best Organizer Award, Seminar on Modern Physics, COMSATS University Islamabad (Aug 2020)

REFERENCES

Dr. Ghulam Hassnain Jaffari — MPhil Supervisor
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